

Classic Tricks

Algebraic Manipulations are a difficult topic to discuss. There is lots of variability within problems you will see and there are by no means a single way to solve them. Still, there are common tricks that you will see which will come up plenty of times in competitions.

- $(x + y, xy)$ or $(x + y + z, xy + yz + zx, xyz)$, etc.
- Summing given equations, subtracting given equations, multiplying given equations, etc. Usually, this simplifies the problem down into something more familiar.
- Substitutions. There are some clever ones: ravi substitution, $\cos(x)$, $\sin(x)$, $\tan(x)$, etc. Trigonometric substitutions are particularly handy and perhaps the problem is hinting at it if you see somethings of the form $\sqrt{1-x}$, $\sqrt{1-x^2}$, $\sqrt{1+x}$, or $\sqrt{1+x^2}$. Familiarly, these are highly reminiscent of the pythagorean identities. There are problems in the lesson that may use this! Sometimes, substitutions just depend on the problem.

Problem 1.1 (1991 AIME). Let x and y be positive integers such that

$$\begin{aligned} xy + x + y &= 71, \\ x^2y + xy^2 &= 880. \end{aligned}$$

Find $x^2 + y^2$.

Problem 1.2 (1983 AIME). What is the product of the real roots of the equation

$$x^2 + 18x + 30 = 2\sqrt{x^2 + 18x + 45}?$$

Common Problems

- Using $x + \frac{1}{x}$ to find $x^n + \frac{1}{x^n}$. Inductively, this is easy since:

$$x^n + \frac{1}{x^n} = \left(x + \frac{1}{x}\right) \left(x^{n-1} + \frac{1}{x^{n-1}}\right) - \left(x^{n-2} + \frac{1}{x^{n-2}}\right)$$

- Given $a + \frac{1}{b}$ and $b + \frac{1}{a}$, find $ab + \frac{1}{ab}$. Time-less problem which are seen throughout the problem sets!

Problem 2.1 (2000 AIME). Given that z is a complex number such that $z + \frac{1}{z} = 2 \cos 3^\circ$, find the least integer that is greater than $z^{2000} + \frac{1}{z^{2000}}$.

Problem 2.2 (2020 AMC). The real number x satisfies the equation $x + \frac{1}{x} = \sqrt{5}$. What is the value of $x^{11} - 7x^7 + x^3$?

Problem 2.3 (2022 PMO). Let a , b , and c be real numbers such that

$$3ab + 2 = 6b, \quad 3bc + 2 = 5c, \quad \text{and} \quad 3ca + 2 = 4a.$$

Find the sum of all possible values of abc .

Factorizations

Many times, solving a problem of this type will boil down to recognizing one or several common factorizations and manipulations. Here are just a few; you will find some others in the exercises.

- $x^2 - y^2 = (x - y)(x + y)$, $x^4 - y^4 = (x - y)(x + y)(x^2 - y^2)$, $x^8 - y^8 = (x - y)(x + y)(x^2 + y^2)(x^4 + y^4)$.
- $x - y$ can be factored out of $x^k - y^k$ for all integer k . More commonly: $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$
- $x + y$ can be factored out of $x^k + y^k$ for all even integer k . More commonly: $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$
- $x^3 + y^3 + z^3 = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx) + 3xyz$. This can also be rewritten using:

$$x^2 + y^2 + z^2 - xy - yz - zx = \frac{1}{2} ((x - y)^2 + (y - z)^2 + (z - x)^2).$$

- $(x + y + z)^3 = x^3 + y^3 + z^3 + 3(x + y)(y + z)(z + x)$
- Less-known, but common: $(x + y)(x + z) = x^2 + xy + yz + zx$.
- [Binomial Theorem](#). This can be useful in number theory problems.

Problem 3.1 (2005 AIME). Define

$$N = \frac{4}{(\sqrt{5} + 1)(\sqrt[4]{5} + 1)(\sqrt[8]{5} + 1)(\sqrt[16]{5} + 1)}.$$

Find $(N + 1)^{48}$.

Problem 3.2 (2021 AMC). Simplify the following expression:

$$(2 + 3)(2^2 + 3^2)(2^4 + 3^4)(2^8 + 3^8)(2^{16} + 3^{16})(2^{32} + 3^{32})(2^{64} + 3^{64}).$$

Problem 3.3 (1983 AIME). Find the remainder when $6^{83} + 8^{83}$ is divided by 49.

Simon's Favorite Factoring Trick

This is a very common trick all mathletes should know.

- Consider the example $2xy - 6x - 7y = 40$ and you wanted to find all integer solutions. To do this, you want to try and factor this as so:

$$4xy - 12x - 14y = 80 \iff (2x - 7)(2y - 6) = 80.$$

Simply go through all divisors of 80 and see which ones work. Occasionally, there are ways to simplify this through divisibility.

- Many problems will have some variation of this. Different numbers can replace 2, 6, 7, and 40. See below!

Problem 4.1 (2021 AMC 12B). Two distinct numbers are selected from the set $\{1, 2, 3, 4, \dots, 36, 37\}$ so that the sum of the remaining 35 numbers is the product of these two numbers. What is the difference of these two numbers?

Problem 4.2 (2018 AMC 10B). How many ordered pairs (a, b) of positive integers satisfy the equation

$$a \cdot b + 63 = 20 \cdot \text{lcm}(a, b) + 12 \cdot \text{gcd}(a, b),$$

where $\text{gcd}(a, b)$ denotes the greatest common divisor of a and b , and $\text{lcm}(a, b)$ denotes their least common multiple?

Problem 4.3 (2008 AMC 10B). A rectangular floor measures a by b feet, where a and b are positive integers and $b > a$. An artist paints a rectangle on the floor with the sides of the rectangle parallel to the floor. The unpainted part of the floor forms a border of width 1 foot around the painted rectangle and occupies half the area of the whole floor. How many possibilities are there for the ordered pair (a, b) ?

Sketches

Some details omitted. Message *jakhocobting* for questions.

Sketch 1.1 Rewrite $a = xy$ and $b = x + y$, then $a + b = 71$ and $ab = 880$. Substitute $a = 71 - b$, obtain a quadratic and solve. Or, as x and y are positive integers, then a and b are integers and find a solution testing divisors of 880. Extract 146.

Sketch 1.2 Substitute $t = x^2 + 18x + 30$, then $t = 2\sqrt{t + 15}$. Square both sides and solve for $t = 5, -3$. Then, solve $t = x^2 + 18x + 30$. Ensure the solutions are real; that is, Vieta's is not allowed. Extract 20.

Sketch 2.2 yawn too lazy, just search it

Sketch 2.2 $x + \frac{1}{x} = \sqrt{5} \implies x^2 + \frac{1}{x^2} = 3 \implies x^4 + \frac{1}{x^4} = 7 \implies x^{11} - 7x^7 + x^3 = \boxed{0.}$

Sketch 2.3 $3a + \frac{2}{b} = 6$, $3b + \frac{2}{c} = 5$ and $3c + \frac{2}{a} = 4$. Multiplying yields a result in terms of abc after subtracting six times the sum of the above fractions. Vieta's finishes.

Sketch 3.1 Let $\sqrt[16]{5} = T$, then $N \cdot (T^8 + 1)(T^4 + 1)(T^2 + 1)(T + 1) = (T^{16} - 1)$. By the first factorization property, then $N = T - 1$. But, $T^{16} = 5$, so $(N + 1)^{48} = T^4 8 = \boxed{125.}$

Sketch 3.2 Multiplying by $(3 - 2)$ yields $3^{128} - 2^{128}$.

Sketch 3.3 Several terms through binomial-theorem on $(7 - 1)^{83}$ and $(7 + 1)^{83}$ vaish modulo 49 yielding a result.

Sketch 4.1 $ab = \frac{35 \cdot 36}{2} - a - b$. SFFT finishes.

Sketch 4.2 Use $\text{lcm}(a, b) \cdot \text{gcd}(a, b) = ab$ to substitute and obtain a form reminiscent of SFFT.

Sketch 4.3 yawn too lazy, just search it